

COMP30850 Sample Exam - Spring 2025/26

Guidelines

- Complete all three tasks. All tasks carry equal marks.
- All tasks should be completed in a single Jupyter Notebook.
- Use Markdown cells to briefly explain your work.
- A time limit of 2 hours applies to this exam. You must submit your Jupyter notebook (IPYNB file) within this time.
- No collaboration is allowed. All submissions will be subject to plagiarism checking. Any evidence of plagiarism can result in a 0 grade.
- This is an open book exam (i.e., you can use module lecture/lab material and internet resources).
- Development environments (such as VSCode or Colab) **must not** be used for the exam. Use web-based Jupyter notebooks to complete the exam.

Use of Generative AI:

- Generative AI tools, code assistants, and large language models (such as ChatGPT, Copilot or Gemini) **must not** be used for any part of this exam.

Overview

This exam involves completing three tasks based on a dataset of financial transactions between customer bank accounts over a three-month period. Each transaction in the dataset includes a date, a source account number, a destination account number, and a transaction amount, representing the flow of funds between two parties.

The raw transaction dataset is available from the link below:

<http://mlg.ucd.ie/modules/COMP30850/transactions.csv>

Task 1: Static Network Creation

- a) Load the CSV file and construct an appropriate static network representation of the financial transactions over the three-month period. Briefly justify your choice of network type and how you handle multiple transactions between the same pair of accounts.

- b) Based on the static network, calculate the following:
 - i. Network density
 - ii. Level of reciprocity in the network
- c) Briefly explain how each metric from Task 1b might be interpreted in the context of a financial transaction network.

Task 2: Static Network Characterisation

- a) Using the static network from Task 1, identify the top 10 accounts based on:
 - i. Weighted in-degree
 - ii. Weighted out-degree
- b) Visualise the full weighted out-degree distribution of the static network.
- c) Briefly explain the difference between *weighted* and *unweighted* out-degree and what each reveals in the context of a financial transaction network.

Task 3: Dynamic Network Creation and Analysis

- a) Using the transaction dataset, construct a dynamic network that captures how the transaction network evolves over the three-month period. Justify your choice of time window size.
- b) Analyse how the dynamic network evolves over time by calculating the following values for each time window network:
 - i. Number of nodes and edges
 - ii. Network density
- c) Based on your results from Task 3b, describe how the network evolves over the three months and briefly suggest possible explanations for any trends you observe.